

Al Amin Wazed Shourov

Ames, Iowa

BSc in Physics, University of Dhaka (DU), Bangladesh

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RESEARCH PROFILE

Prospective PhD applicant with a B.Sc. in Physics and research experience across computational condensed matter physics, DFT-based materials simulation, liquid-metal surface tension modeling, nonlinear dynamics, soft-matter self-propulsion, topological properties, and molecular dynamics. Interested in **theoretical, computational, and experimental condensed matter physics, as well as materials science**, with a focus on connecting physical mechanisms, numerical modeling, experimental characterization, materials synthesis, measurement, and interpretable materials or dynamical-system behavior.

EDUCATION

Bachelor of Science in Physics

Jan 2020 – June 2025

University of Dhaka, Dhaka, Bangladesh

CGPA: 3.10/4.00

Higher Secondary School Certificate

2016 – 2018

Notre Dame College, Dhaka, Bangladesh

GPA: 5.00/5.00

RESEARCH INTERESTS

- Computational Condensed Matter physics
- Experimental Condensed Matter Physics
- material science and soft matter physics
- molecular dynamics
- Nonlinear Dynamics

TEST SCORES

GRE Physics Test: 850 (86% in CM, 77% in EM)

IELTS Academic: 7.0 (Listening 8.0, Reading 7.5, Writing 6.0, Speaking 6.5)

RESEARCH EXPERIENCE

Condensed Matter Physics Research Group | Department of Physics, University of Dhaka

Supervisor: Dr. Ratan Chandra Gosh, Professor, Department of Physics, University of Dhaka

Surface Tension of 3d Liquid Metals: A Role of Structure Dependent Part

Mar 2024 - Jun 2025

- Studied the microscopic origin of surface tension in liquid transition metals using pseudopotential-based numerical modeling and liquid-state structural information.
- Implemented the *Bretonnet-Silbert* pseudopotential approach and analyzed how model potentials, structure factors, and local-field correction influence calculated surface properties.
- Demonstrated methodology in numerical simulation, parameter handling, comparison with literature, experimental trends, and physical interpretation of why transition-metal bonding makes simple-liquid modeling difficult.
- Presented related work at the **National Conference on Physics 2025** organized by the Bangladesh Physical Society.
- currently under revision following reviewer comments at *Surfaces and Interfaces (IF 6.0)*

Multiscale Simulation of Source–Sink Chemistry and Marangoni Propulsion in Self-Propelled Droplet Robots

April 2026 - Present

- Studying viscosity, transport behavior, and soft-matter response in polymer-surfactant systems using LAMMPS-based simulation planning.
- Implemented conceptual understanding of Marangoni-driven motion, surface-tension gradients, source-sink chemical fields, and how local concentration gradients can generate directed motion in active or soft-matter systems.

- Processing measurable simulation outputs including velocity, diffusion, concentration profiles, viscosity/CAC-related behavior, stability of motion, and parameter dependence of self-propulsion.

Computational Physics Research Group | Department of Physics, University of Dhaka

Supervisor: Dr. Alamgir Kabir, Professor, Department of Physics, University of Dhaka

First-Principles Design of Lead-Free Double Perovskites for Optoelectronic, Mechanical, and Thermoelectric Applications **April 2026 – Present**

- Developed knowledge of how structural relaxation, magnetic ordering, electronic band structure, density of states, optical response, mechanical constants, charge analysis, and thermoelectric indicators connect to possible optoelectronic and energy-related applications.
- Demonstrated DFT workflow design: crystal-structure preparation, pseudopotential selection, convergence of k-point mesh and energy cutoff, and comparison of magnetic configurations in VASP and Quantum ESPRESSO-style calculations.

Nonlinear Dynamics Research Group | Department of Physics, University of Dhaka

Supervisor: Prof. Golam Dastegir Al-Quaderi, Professor, Department of Physics, University of Dhaka

Variable-Dependent Quantification of Chaos in the Nonlinear System **Feb 2025 - Nov 2025**

- Worked with chaotic systems to understand stability, sensitivity to initial conditions, phase-space structure, and how complexity measures distinguish periodic, quasi-periodic, and chaotic behavior.
- Calculated chaotic attractors and their characteristics using numerical integration, trajectory visualization, phase portraits, and parameter sweeps.
- Computed entropy-based measures and autocorrelation analysis to investigate system complexity and temporal memory; gained awareness that entropy metrics require careful treatment of binning, normalization, and comparison with other chaos indicators such as Lyapunov exponents.

TECHNICAL SKILLS

Programming and Scientific Computing: Python, MATLAB, C/C++, FORTRAN; NumPy, SciPy, Matplotlib, Astropy, Lightcurves.

DFT and Atomistic Simulation and Structure: VASP, Quantum ESPRESSO, LAMMPS, VESTA, Avogadro, SCAPS-1D

Numerical Methods and Analysis: pseudopotential modeling, nonlinear ODE simulation, scientific plotting, result interpretation, Gnuplot, Origin Pro

Documentation and General Tools: LaTeX, MS Word, PowerPoint, Excel.

Experimental Experience: 4 years of academic laboratory experience in electronics, nuclear physics, solid state physics, optics and waves, mechanics, thermodynamics, electricity and magnetism, and modern physics.

AWARDS AND ACHIEVEMENTS

- Ranked 5th in Dhaka Division in Secondary Exam 2016 and was nominated for a two-year scholarship.
- Ranked 16th in Dhaka Division in Higher Secondary Exam 2018 and was nominated for a four-year scholarship.
- Won Division round of Exploring Creative Talent Competition 2015 in the Math and Computer domain.
- Won National round of General Knowledge Olympiad in 2015 and 2012.
- Won Division round of BAS Science Olympiad 2016.

SERVICE AND EXTRACURRICULAR ACTIVITIES

- Executive panel member, Notre Dame Nature Study Club (NDNC), 2017-2018.
- Volunteer, Physics Olympiad organized by BdPhO.
- Volunteer, Centennial Celebration of Bose-Einstein Statistics: A Legacy to Dhaka, Department of Physics, University of Dhaka.
- Volunteer, National Conference on Physics 2025 organized by the Bangladesh Physical Society.

REFERENCES

Dr. Ratan Chandra Gosh, Professor, Department of Physics, University of Dhaka | Email: ratan@du.ac.bd

Dr. Alamgir Kabir, Professor, Department of Physics, University of Dhaka | Email: alamgir.kabir@du.ac.bd

Prof. Golam Dastegir Al-Quaderi, Professor, Department of Physics, University of Dhaka | Email: dastegir@du.ac.bd